

Part 2 - Prediction System

Group 9

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This part focuses on building a prediction system to assess student academic success. We utilize a total of 9 features, integrating the actionable factors from Part 1 supplemented with study environment data. The output is formulated as a regression task for predicting final scores and a classification task for pass/fail status. For data preprocessing, we normalize all numerical features and apply one-hot encoding to categorical variables.

1 Selected Models

We employ classical machine learning models as the backbone of our prediction system. We frame KNN as a classification task due to poor regression performance, while utilizing a regression framework for the tree-based models:

- **KNN:** A non-parametric method that predicts the class of a new data point based on the majority vote of its k closest examples in the training set.
- **Random Forest:** An ensemble learning method that constructs multiple decision trees via bootstrapping during training. It aggregates their outputs to improve overall accuracy and mitigate overfitting.
- **Gradient Boosting:** A sequential ensemble technique that iteratively adds weak learners (typically decision trees) to correct the residual errors of previous models, optimizing a predefined loss function via gradient descent.
- **XGBoost:** An optimized implementation of gradient boosting that uses a second-order Taylor expansion to approximate the loss function and incorporates regularization to control model complexity and prevent overfitting.

2 Training Strategy

Following the procedures outlined in Part 1, we partitioned the dataset into training and testing sets using an 80:20 split. We employed `RandomizedSearchCV`

for hyperparameter optimization during training. Model selection was evaluated using negative Mean Absolute Error (MAE) for the regression models and accuracy for the classification model.

3 Model Comparison

The performance of the selected models was evaluated using classification metrics (Accuracy, F1-Score, AUC-ROC) and regression metrics (Mean Absolute Error).

- **Classification Accuracy:** Random Forest achieved the highest Accuracy (0.940), followed closely by XGBoost (0.935).
- **Regression Precision:** XGBoost demonstrated strong regression performance with a low MAE of 5.82, significantly better than Random Forest's 6.02.
- **Distinguishing Ability:** XGBoost tied for the highest AUC-ROC at 0.95, indicating superior reliability in distinguishing between passing and failing students compared to KNN (0.50) and Random Forest (0.94).
- **Model Selection:** Based on its ability to minimize regression error while maintaining high classification confidence, XGBoost was selected as the optimal model for this prediction system.

4 SHAP Explanation

4.1 Methodology and Operation

SHAP (SHapley Additive exPlanations) is a game-theoretic approach used to explain the output of machine learning models. For our XGBoost regressor, we utilized the TreeExplainer, which is specifically optimized for tree-based ensembles.

- **Core Concept:** SHAP assigns each feature a "Shapley value" for a particular prediction, measuring its contribution to the difference between the actual prediction $f(x)$ and the average predicted value $E[f(X)]$.
- **Calculation:** The algorithm ensures a fair distribution of "credit" among input variables by considering all possible feature combinations.

4.2 Output and Interpretation

The primary output is the **Waterfall Plot**, which visualizes the decision-making process for an individual student.

- **Base Value ($E[f(X)]$):** The average predicted score for the dataset is 71.06.

- **Positive Drivers (Red):** Features like physical activity minutes (+0.95), homework completion (+0.41), and daily study hours (+0.33) pushed the score higher.
- **Negative Drivers (Blue):** **Attendance percentage** was the most influential factor, drastically reducing the prediction by 12.96 points. Screen time (-0.74) and motivation (-0.69) also had negative impacts.
- **Final Prediction ($f(x)$):** The sum of these contributions resulted in a final score of 58.635.

5 Overall Conclusion

This part developed a comprehensive prediction system by integrating actionable academic and lifestyle factors to assess student success.

- **Model Selection Process:** Among the four candidate models (KNN, Random Forest, Gradient Boosting, and XGBoost), **XGBoost** was identified as the optimal choice. While Random Forest showed slightly higher classification accuracy (0.940), XGBoost provided a superior balance with a significantly lower Mean Absolute Error (5.82) and the highest tied AUC-ROC (0.95).
- **Interpretability via SHAP:** To move beyond "black-box" predictions, we applied SHAP explanation specifically to the XGBoost model. This allowed for a granular decomposition of feature contributions, revealing how individual factors like attendance and physical activity shift the predicted final score relative to the population average.
- **Final Recommendation:** We recommend the deployment of the XGBoost-based system for its predictive precision and transparency. Future interventions should focus on data-driven strategies to improve attendance and promote physical activity to effectively boost student outcomes.